Multimodal AI Assistant for Smart Agriculture

Milestone 6

Crop Disease Detection + RAG Advisory System
for Indian Farmers

Group 8 - IITM - DSAI Lab - 2026

Our Team

Harish Sahadev M

Sai Naman

Allanki Saketh Kumar

Manas Rastogi

Ayushi Dixit

Problem Context

Farmers in India face major challenges in timely crop disease detection and accessing reliable agricultural guidance. These challenges are intensified by language barriers, limited digital access, and shortage of expert support in rural areas.

- Crop diseases destroy **20–40% of global crop yields annually**
- Indian farmers lack **quick, reliable disease identification tools**
- **Agricultural experts are scarce** in rural regions
- **Language barriers** limit access to digital advisories

Proposed Solution

A **Multimodal AI Assistant for Smart Agriculture** that integrates computer vision, generative AI, and speech technologies to deliver accessible, practical support tailored to Indian farming conditions:

- Detects crop diseases from **leaf images** (rice, wheat, sugarcane, potato, corn)
- Supports **multilingual chatbot** with **voice & text interaction**
- Uses **RAG** for grounded, reliable recommendations
- Designed for **low-bandwidth rural environments**

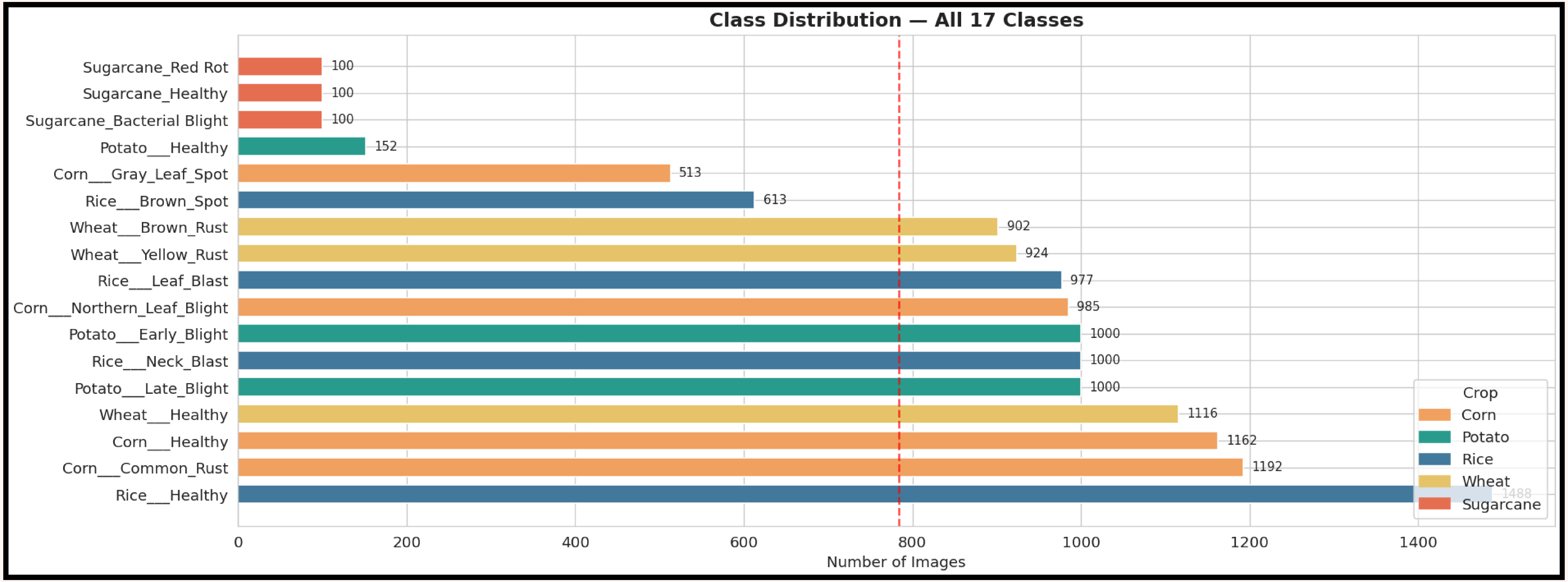
Goal: Provide an **accessible, accurate, and context-aware** AI assistant for farmers.

Dataset Overview

The project uses the **Top Agriculture Crop Disease dataset**, comprising labeled leaf images across multiple crops and disease conditions. The dataset is suitable for supervised learning and captures **real-world variability** such as lighting, background noise, and leaf damage.

- **Total Images:** 13,324
- **Total Classes:** 17 (disease + healthy)
- **5 Crops Species:** Rice, Wheat, Sugarcane, Potato, Corn
- **Data Split:** Train (80%) | Val (10%) | Test (10%)
- **Image Resolution:** Resized to 224 × 224

Dataset Overview (cont.)



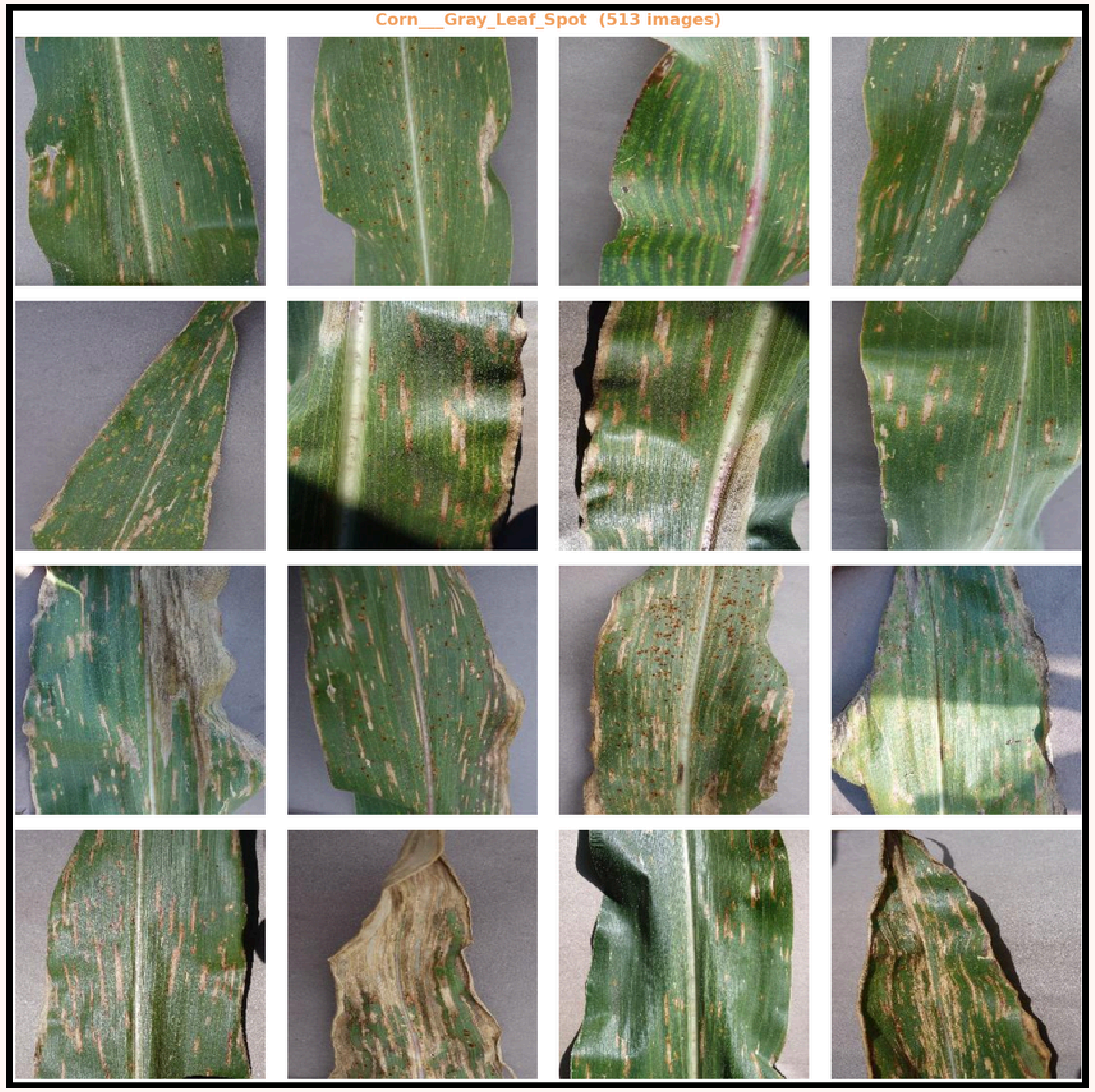
Sugarcane severly under-represented - addressed via target augmentation

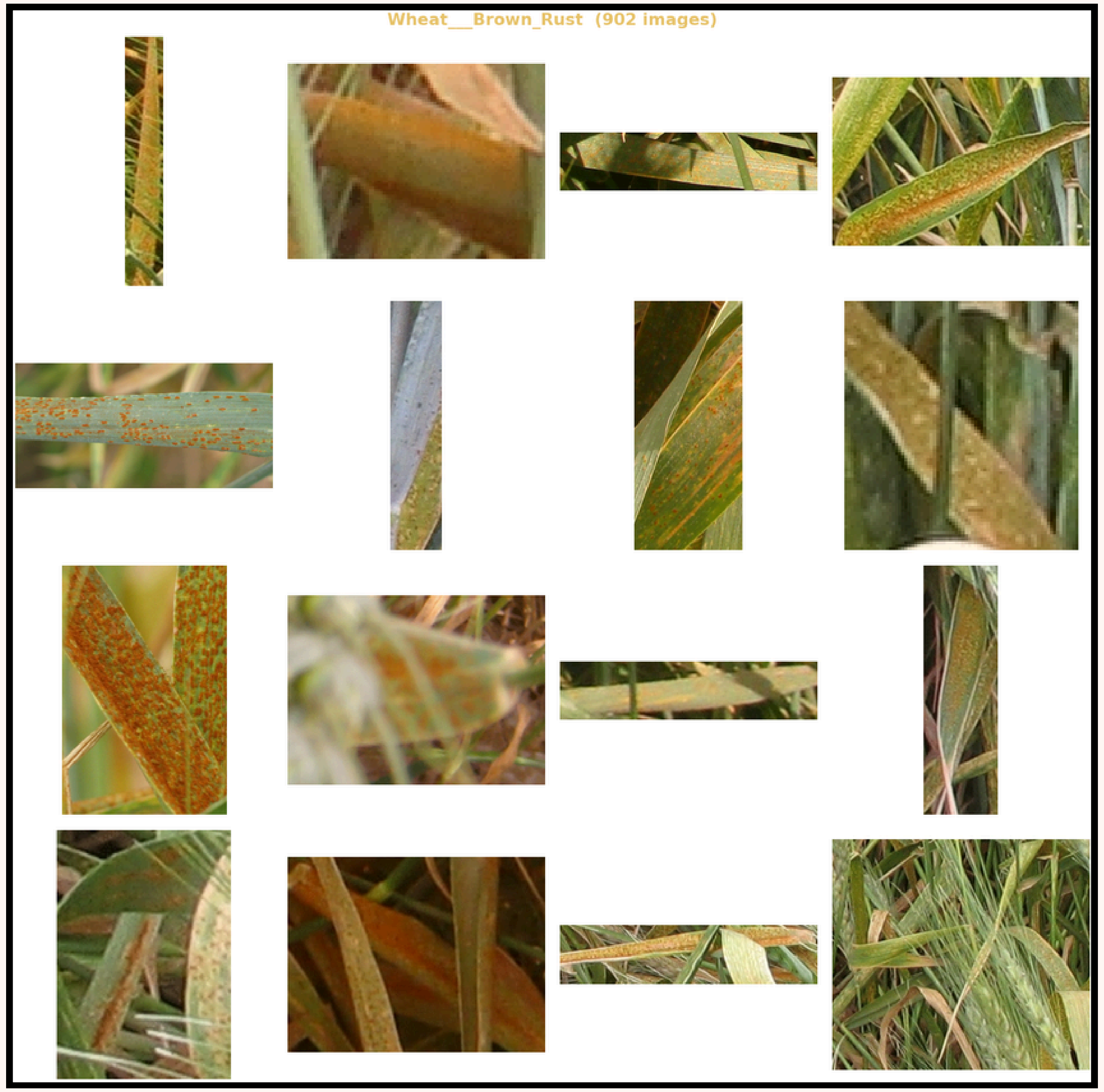
Dataset Characteristics and Insights

The dataset contains diverse leaf images with variations in lighting, background, and leaf damage, making it suitable for real-world deployment scenarios.

- Includes both **healthy and diseased** samples per crop
- Covers **major diseases** like blight, rust, blast, etc.
- Shows class imbalance across categories
- Images vary in resolution and quality

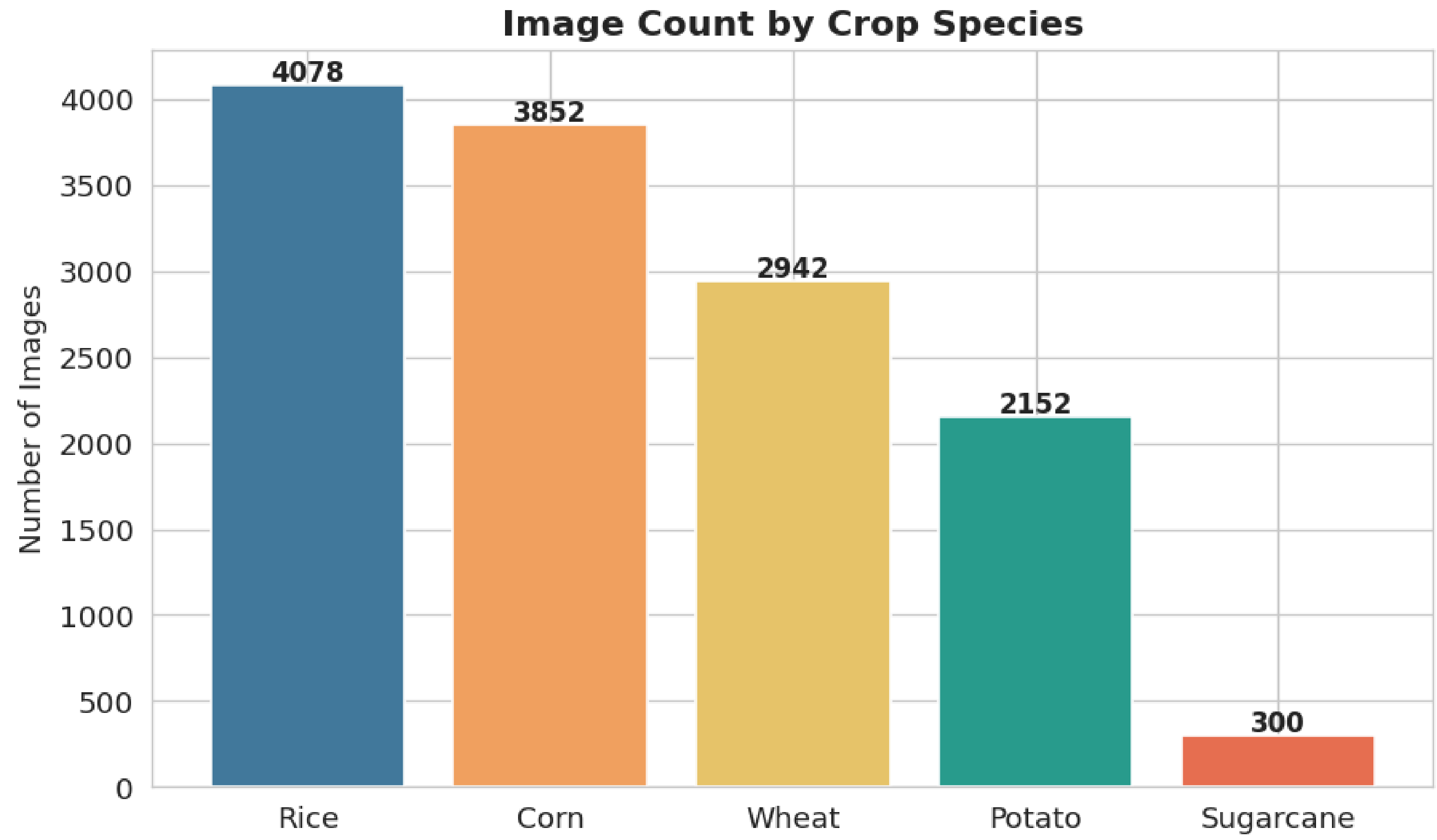
Dataset Examples





EDA - Crop Distribution and Image Resolution

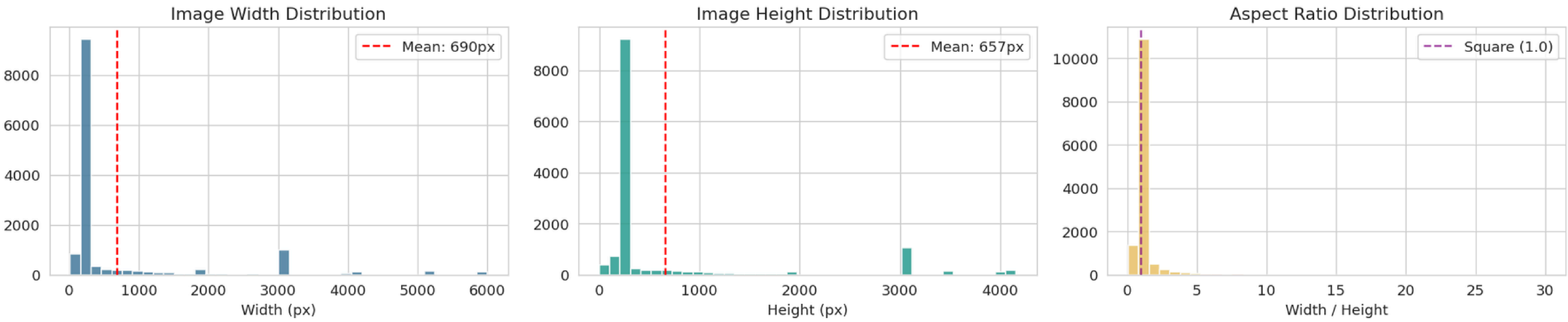
Crop-Level Distribution



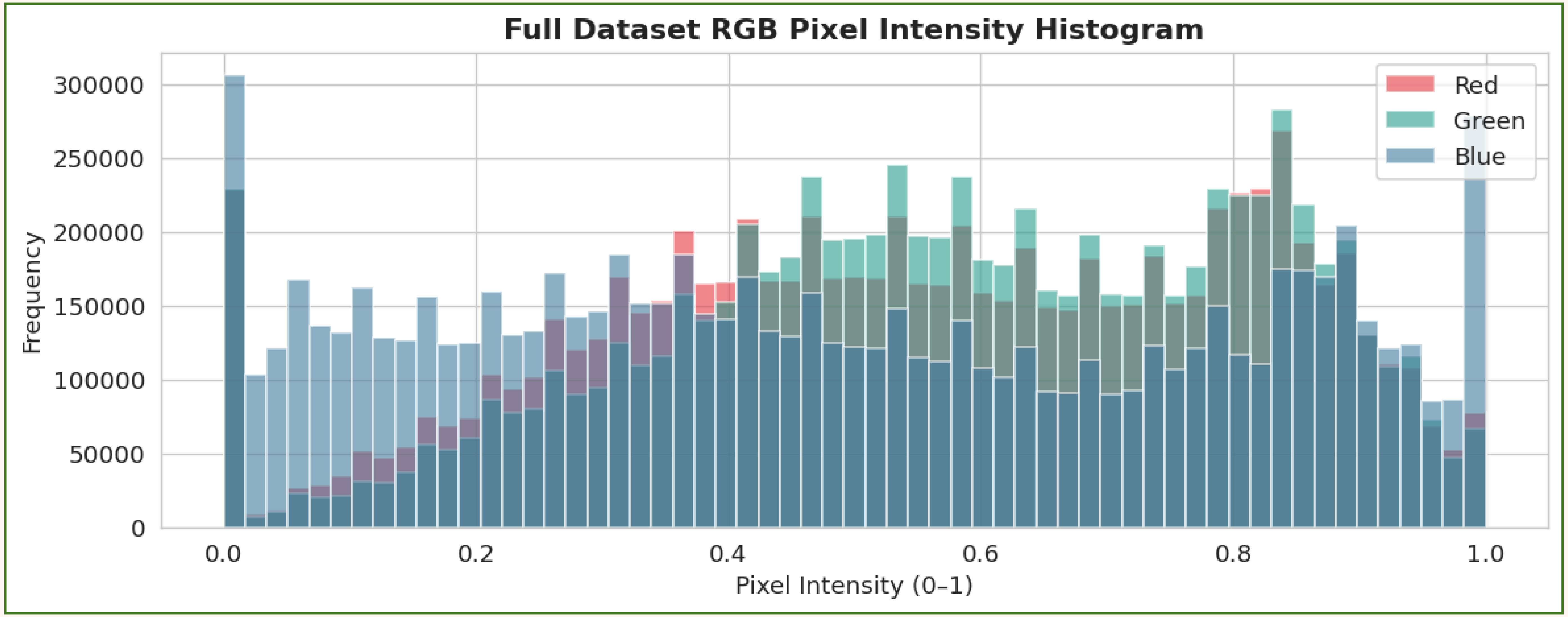
Key EDA Findings

Rice dominates:	30.6% of all images
Sugarcane tiny:	Only 2.3% of dataset
All RGB:	0 non-RGB images found
Width range:	16px – 6,000px (mean 690px)
Height range:	1px – 4,160px (mean 657px)
Near-dupes:	5 pairs found in 500-sample
Exact dupes:	0 found (MD5 clean)

Image Size Analysis



EDA - Pixel Intensity and Color Analysis



Color Channel Statistics

Channel	Mean	Std Dev
Red	0.5576	0.2481
Green	0.5783	0.2378
Blue	0.4862	0.2990

Key Colour Insights

- Green channel slightly dominant (mean 0.578) — consistent with leaf chlorophyll content
- Blue channel lowest (mean 0.486) — less blue reflectance from vegetation
- Bimodal distributions visible — mix of dark soil backgrounds and bright leaf surfaces
- All channels sampled from 500 images with random_state=42 for reproducibility

PREPROCESSING PIPELINE

1	2	3	4	5	6
Resize 224×224	RGB Conversion	Duplicate Removal	Augmentation	Normalisation	Stratified Split

1. Exact Deduplication (MD5)

MD5 hash computed for all 13,324 images. Result: 0 exact duplicates found. Dataset integrity confirmed.

2. Near-Duplicate Removal (pHash)

Perceptual hash (pHash) with Hamming distance ≤8. EDA sample found 5 near-duplicate pairs. Full-dataset removal applied in preprocessing.

3. LANCZOS 224×224 Resize

All images resized to 224×224px using LANCZOS (high-quality downsampling). Chosen over bilinear for sharper leaf texture preservation.

4. RGB Verification

Channel check confirms all 13,324 images are RGB. Non-RGB images (greyscale, RGBA) would cause training errors — 0 found.

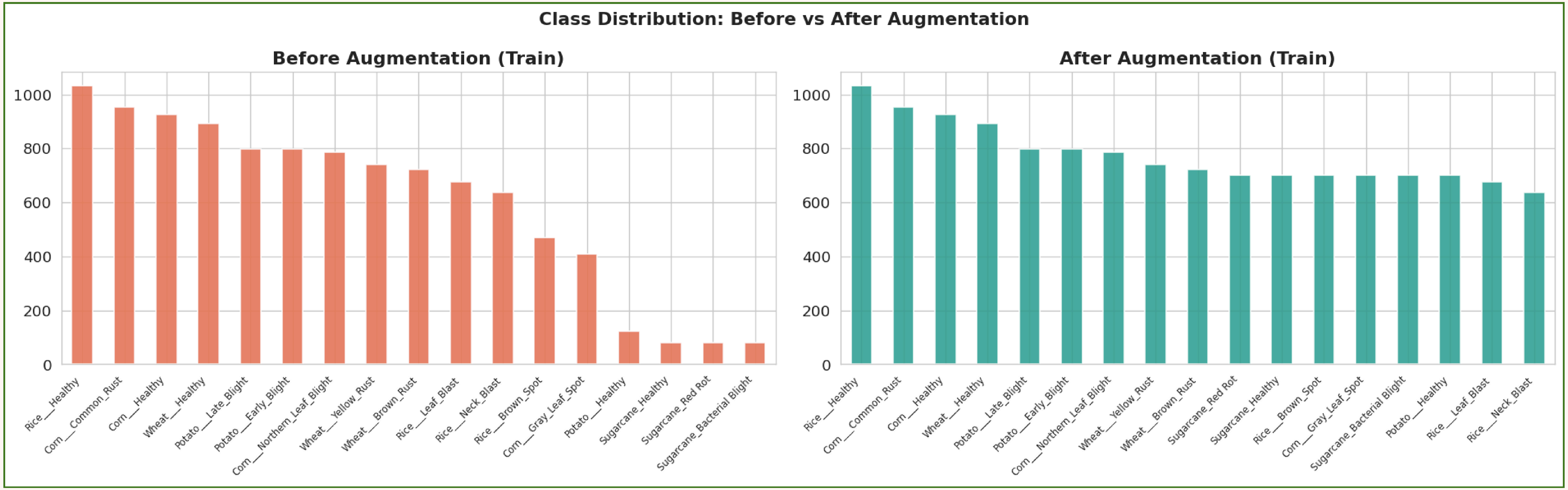
5. Stratified 80/10/10 Split

Train: 10,203 | Val: 1,275 | Test: 1,276 (seed=42). All 17 classes guaranteed in every split. Prevents class-missing folds.

6. Minority Augmentation → 700

6 classes with <600 training images augmented: flip, rotate 0/90/180/270°, brightness ±30%, contrast ±20%, saturation ±20%, zoom ±15%.

Augmentaion Strategy - Fixing Class Imbalance



Augmentation Operation:

- Horizontal Flip
- Vertical Flip
- Rotate 0°/90°/180°/270°
- Brightness ±30%
- Contrast ±20%
- Saturation ±20%
- Zoom crop ±15%

Final Dataset:

- Train 13,161 (incl +2958 augmented)
- Val 1,275
- Test 1,276
- Total - 15,712 images
- Seed = 42
- No leakage

Model Architecture and Comparision

Input Image (N,3,224,224)

↓

Pretrained Backbone — Frozen

↓

Squeeze-and-Excitation Blocks

↓

AdaptiveAvgPool → Flatten (N,960)

↓

Linear(960→1280) + Hardswish

↓

Dropout(p=0.4704) ← Optuna

↓

Linear(1280→17) ← 17 classes

↓

Softmax → (label, confidence)

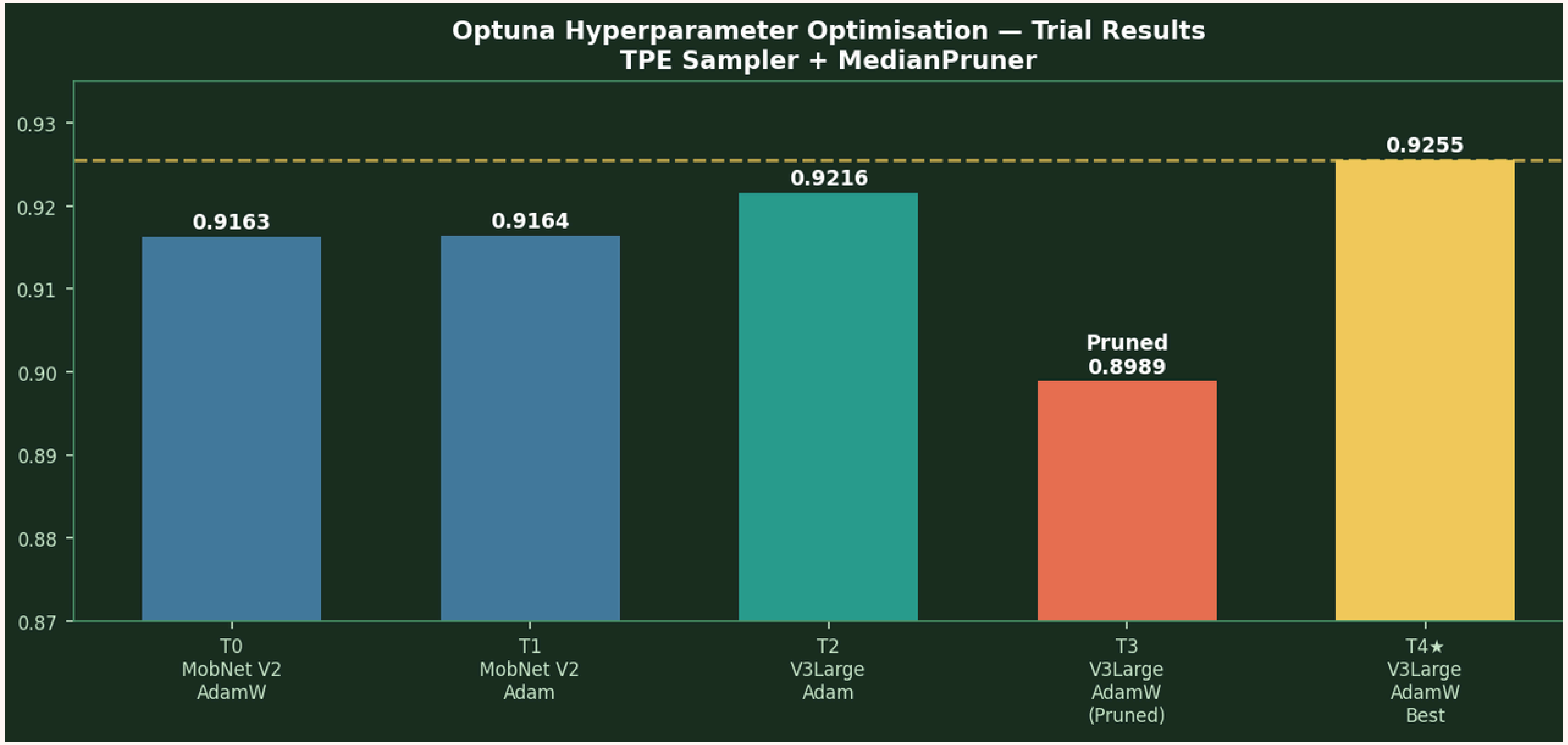
Architecture Comparison

Model	Params	Strategy
MobileNet V3L ★	5.4M	Freeze backbone
MobileNet V2	3.4M	Freeze backbone
ResNet-50	25.6M	Freeze layers 1–3
EfficientNet-B3	12.2M	Freeze backbone

Key Architecture Features

Depthwise Separable Conv	Reduces computation ~8-9× vs standard conv — CPU-friendly
Squeeze-and-Excitation	Channel-wise attention — recalibrates leaf disease feature maps
Hardswish activation	Faster than Swish, matches accuracy — modern mobile efficiency
Frozen backbone	Prevents catastrophic forgetting of ImageNet edge/texture features
17-class classifier head	Only classifier trained — 1.5M trainable params out of 5.4M

Hyperparameter Optimisation - Optuna TPE



★ Best Config (Trial 4)

Architecture	MobileNet V3 Large
Optimizer	AdamW
Learning Rate	4.28e-4
Dropout	0.4704
Weight Decay	1.8e-6
Batch Size	16
Val Macro F1	0.9255 (92.55%)

Search Space & Methodology

Hyperparameter	Range	Sampling
Architecture	mobilenet_v2 / v3_large	Categorical
Optimizer	Adam / AdamW / SGD	Categorical
Learning Rate	1e-5 – 1e-3	Log-uniform
Dropout	0.0 – 0.5	Float uniform
Weight Decay	1e-6 – 1e-3	Log-uniform
Batch Size	16, 32	Categorical

Why Optuna TPE + MedianPruner?

- TPE builds probabilistic model of good hyperparameter regions — smarter than grid/random search
- MedianPruner kills trials at epoch E if val F1 below median — saves compute
- Max 10 epochs per trial — fast iteration on Colab T4 GPU

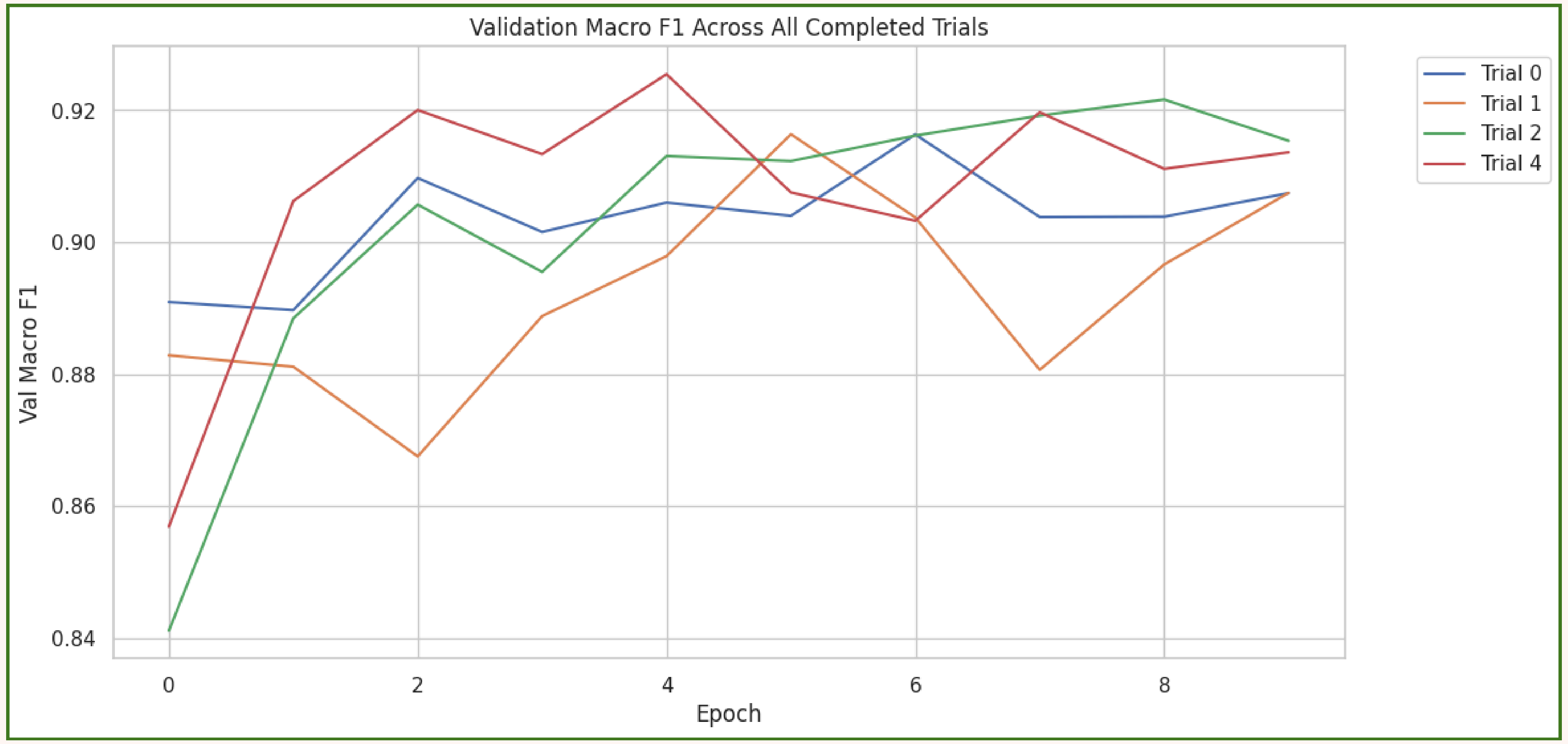
Mobilenet V3 Results and Performance

92.1%
Test Accuracy

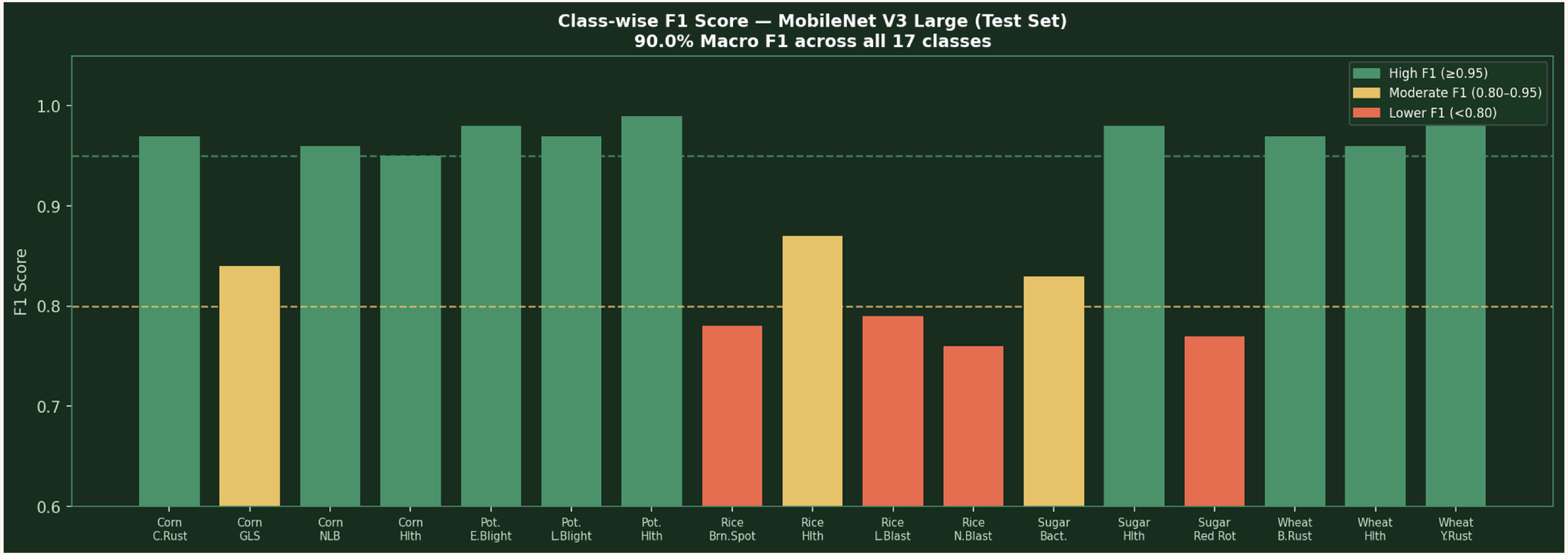
90.0%
Test Macro F1

0.207
Test Loss

92.5%
Val Macro F1



Class-wise F1 Score Analysis



High F1 (≥0.95)

- Corn Common Rust (0.97)
- Corn Northern Leaf Blight (0.96)
- Potato Early/Late Blight (0.97–0.98)
- Sugarcane Healthy (0.98)
- Wheat Brown/Yellow Rust (0.97–0.98)

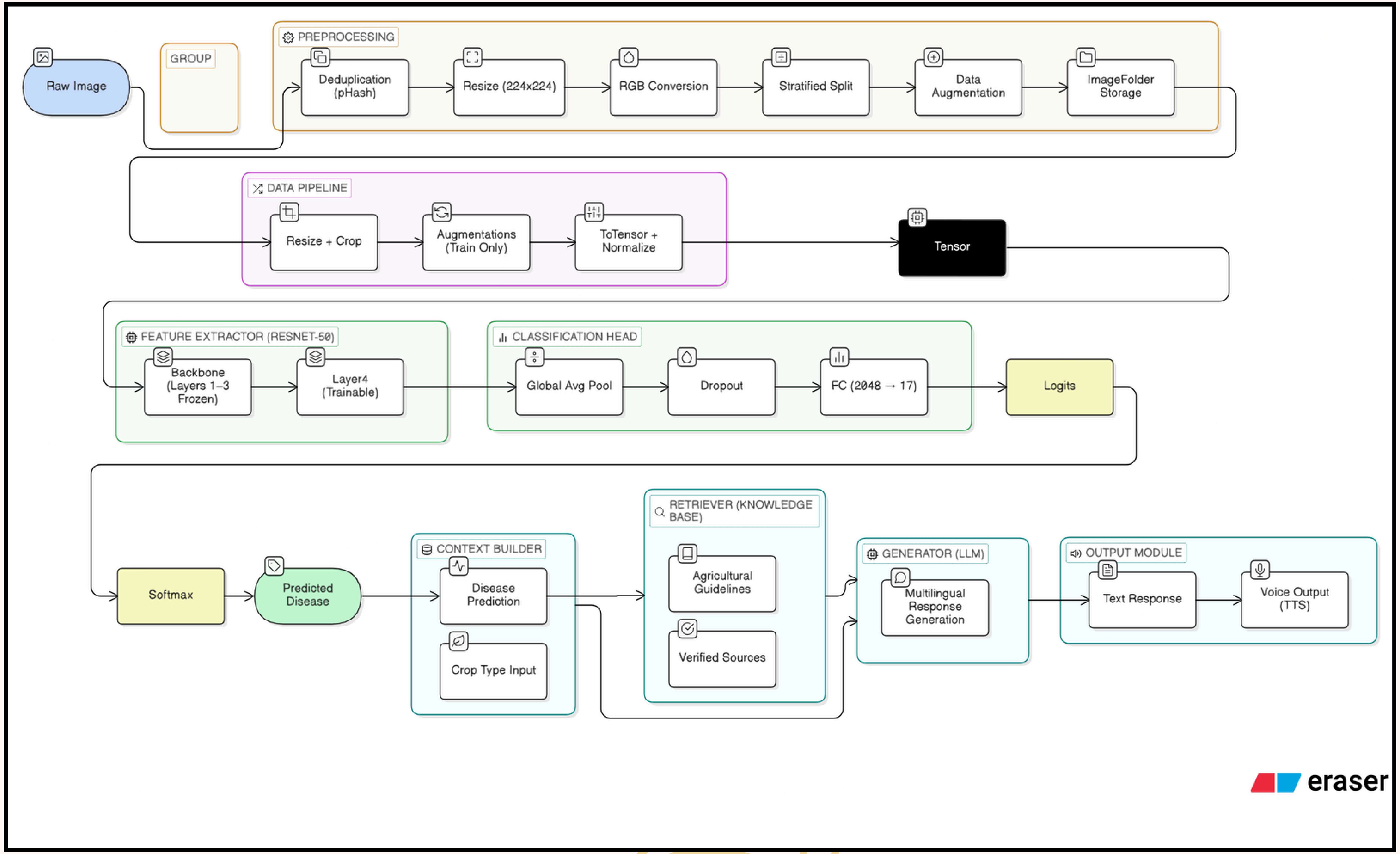
Moderate F1 (0.80–0.95)

- Corn Gray Leaf Spot (0.84)
- Rice Healthy (0.87)
- Sugarcane Bacterial Blight (0.83)

Lower F1 (<0.80)

- Rice Brown Spot (0.78)
- Rice Leaf Blast (0.79)
- Rice Neck Blast (0.76)
- Sugarcane Red Rot (0.77)

System Architecture



End-to-End Application Pipeline

RAG Pipeline – Agricultural Advisory

Why RAG over Pure LLM?

Pure LLMs hallucinate chemical names, wrong dosages, and treatments inapplicable to Indian conditions. RAG grounds every answer in specific verified ICAR/TNAU/FAO documents. If it's not in the knowledge base, the system says so.

Knowledge Base Sources

ICAR — Indian Council of Agricultural Research: disease management, field advisories
TNAU Agritech Portal — disease identification, chemical control schedules
FAO — Food & Agriculture Organisation: crop protection manuals
Naming: {crop}_{disease}_{source}.pdf → auto metadata parsing | 34 chunks indexed

3-Stage Fallback Retrieval Strategy

Stage 1

crop == 'corn' AND disease == 'gray leaf spot'
+ semantic cosine similarity ranking
Top 4 chunks returned

→

Stage 2 (if < 2 results)

crop == 'corn' only
+ semantic cosine similarity ranking
Broader crop-level context

→

Stage 3 (if < 2 results)

No metadata filter
Pure semantic search across all 34 chunks
Last resort — cross-disease retrieval

Embedding: intfloat/multilingual-e5-large

560M params | 1024-dim | 100+ languages | Cosine similarity
Farmer query in Tamil → directly matches English agricultural documents (no translation step)

Confidence Threshold (60%)

≥60% → full crop+disease filter pipeline
<60% → crop-level only + LLM cautious response + KVK referral

Voice Interface ; 7-Language Support

Hindi	Bengali	Tamil	Telugu
hi • 400M+ speakers	bn • 230M+	ta • 80M+	te • 85M+

Malayalam	Kannada	English
ml • 38M+	kn • 44M+	en • Official

Whisper ASR (Input)

- Model: openai/whisper-small — 244M parameters
- Auto language detection from audio — no dropdown needed
- Audio → Mel spectrogram → language ID → transcription
- Chosen over whisper-medium/large to fit 16GB RAM budget
- Language code flows downstream: retrieval, LLM, TTS all respond in detected language

gTTS Text-to-Speech (Output)

- Google TTS API wrapper — free, no API key required
- All 7 languages natively supported
- Text capped at 1,500 chars ≈ 1 min audio at natural rate
- MP3 saved to /tmp/ → returned to Gradio audio player
- Fallback to English TTS if target language fails

Key Design Decisions and Trade-offs

Why MobileNet V3L over ResNet-50?

25.6M vs 5.4M params. MobileNet ~5× faster on CPU. Achieved 92.1% vs ResNet ~91.2% — better accuracy AND speed. Rural deployment target = CPU-only inference.

Why Macro F1 as HPO objective?

14.9× imbalance makes accuracy misleading. A model ignoring Sugarcane (2.3% of data) could hit 85% accuracy. Macro F1 forces equal weight on all 17 classes.

Why RAG over pure LLM?

LLMs hallucinate chemical names, wrong dosages, non-Indian treatments. RAG pins every answer to a specific ICAR/TNAU/FAO document. No verification = dangerous advice.

Why multilingual-e5-large?

English-only embedding models give near-random cosine similarity for Tamil or Telugu queries. e5-large trained on 100+ parallel corpora — Indian language queries directly match English documents.

Why Groq over OpenAI?

Groq free tier provides <1s Llama 3.3 70B inference. OpenAI GPT-4o: paid ($0.01–0.03/call). HF Inference: 5–30s latency. Zero cost + production latency = Groq.

Why whisper-small over medium/large?

16GB RAM budget on HF Spaces free tier shared with MobileNet+e5-large. whisper-small (244M params, ~950MB) fits; whisper-large (1.55B, ~3.1GB) would exhaust RAM.

Challenges Faced and Resolutions

⚠ 14.9× class imbalance (Sugarcane: 80 train imgs)

Targeted augmentation → 700 training images. Macro F1 objective forces equal optimisation pressure. Result: Macro F1 from ~70% → 90%.

⚠ Transform mismatch: training vs inference pixel distribution

M2 preprocessed to LANCZOS 224×224 directly. inference.py aligned to same pipeline — bypasses standard Resize(256)→CenterCrop(224) convention.

⚠ Cross-crop retrieval confusion (rust disease vocabulary overlap)

3-stage metadata pre-filtering in ChromaDB. crop+disease → crop-only → unfiltered. Pins retrieval to correct disease category before semantic ranking.

⚠ Non-English queries → English documents (no translation)

multilingual-e5-large (560M, 100+ languages) maps Indian language queries and English text into the same embedding space. No translation API needed.

⚠ 16GB RAM budget: MobileNet + Whisper + e5-large

whisper-small (244M) chosen over medium/large. Total: MobileNet ~80MB + e5-large ~2.1GB + whisper ~950MB + overhead ≈ 8-10GB. Fits comfortably.

⚠ Confident wrong predictions on blurry/non-leaf images

60% confidence threshold → low_confidence flag → crop-level retrieval only + LLM cautious response + KVK field consultation recommendation.

Deployment Architecture

Infrastructure Stack

Component	Service	Cost
App hosting	Hugging Face Spaces	Free
LLM inference	Groq API (Llama 3.3 70B)	Free
Vector store	ChromaDB (local)	Free
ASR	Whisper-small (local)	Free
TTS	gTTS (Google)	Free
CI/CD	GitHub → HF Spaces push	Free
Model storage	HF Spaces file browser	Free

HF Spaces Free Tier Constraints

RAM	**16GB**	MobileNet+Whisper+e5 ≈8-10GB
CPU	**2 vCPU**	All inference on CPU
Storage	**50GB**	Model+ChromaDB+code
GPU	**None**	CPU-only inference
Latency	**~3-8s**	End-to-end per request

Deployment Steps

1. Create HF Space (Gradio SDK 6.10.0)
2. Set GROQ_API_KEY in Space Secrets
3. Upload mobilenet.pth via file browser → model/
4. Upload rag/chroma_db/ via file browser
5. git push hf main --force to trigger auto-build
6. Monitor build logs → Space live at *.hf.space

Limitations and Future Roadmap

Current Limitations

⚠ 34 doc chunks
Narrow advisory — needs more ICAR/KVK documents

⚠ whisper-small
Heavy accents → occasional transcription errors

⚠ gTTS needs internet
Cannot work offline in no-connectivity areas

⚠ CPU-only inference
~3-8s end-to-end latency per request

⚠ No severity output
Cannot classify mild vs moderate vs severe disease

⚠ 5 crops only
Cotton, groundnut, soybean not yet covered

Future Roadmap

Disease Severity
Mild/moderate/severe output head on classifier

Larger Knowledge Base
State agri uni bulletins, 200+ chunks

Multimodal RAG
LLaVA/Gemini to caption PDF figures

Offline Mode
Llama 3.2 3B GGUF + FAISS — rural deployment

More Crops
Cotton, soybean, groundnut, tomato

Severity-aware Retrieval
Tag chunks by severity; filter at query time

Key Takaways

01

Right metric shapes everything

Choosing Macro F1 as the training objective — not accuracy — directly drove minority-class performance. Sugarcane went from ~70% F1 to 90%. Metric choice is a design decision.

02

Augmentation can close a 15× gap

100 original Sugarcane images → 700 training images via targeted augmentation. Combined with Macro F1 objective, this erased what could have been a catastrophic performance gap.

03

RAG is non-negotiable in high-stakes domains

A pure LLM would confidently recommend wrong fungicides. RAG ensures every recommendation traces to a specific ICAR/TNAU page. Correctness > fluency.

04

Multilingual embeddings eliminate translation

One e5-large model handles Indian languages querying English documents. No translation API, no added latency, no dependency.

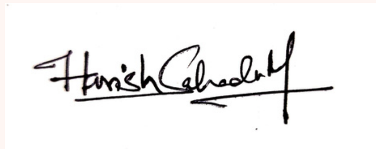
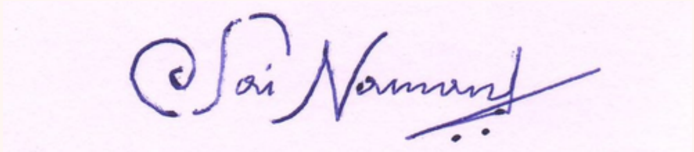
05

Production-grade AI at zero cost is real

HF Spaces + Groq + ChromaDB + gTTS + Whisper = a complete live advisory system for Indian farmers — deployed at $0. Resource constraints drive creative engineering.

Thank You

Live App: https://huggingface.co/spaces/harishsahadev/crop-disease-assistant
Github Repository: https://github.com/22f3000350/Group-8-DS-and-AI-Lab-Project

Harish Sahadev M	**21F1005856**	
Sai Naman	**22F3000350**	
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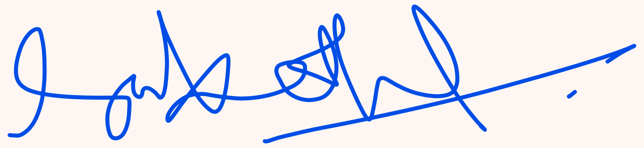
Allanki Saketh Kumar	**21F3002277**	
Manas Rastogi	**22F3001477**